Tufts University, Gordon Institute
TML193-08 - Special Topics: Data & ML for Financial Markets
Spring Semester 2026

Welcome to TML193-08! I am looking forward to meeting you. This syllabus contains important information that will help you understand the expectations and schedule for the course. Thank you for taking the time to read the syllabus thoroughly. I encourage you to refer to it throughout the semester.

COURSE INFORMATION

Meeting Time:	Wednesdays, 6:00–9:00 pm
Meeting Location:	Eaton Hall, Room 225
Instructor:	Josep Puig Ruiz, Lecturer [Redacted for privacy]
Instructor Office Hours:	Eaton Hall, Room 221, Wednesdays, 5:00—6:00pm; email me beforehand if you plan to attend
Semester Hour Units:	Three credits

COURSE DESCRIPTION

This course equips students with the practical skills and domain knowledge required to apply data science and machine learning techniques in capital markets. Students will learn to handle financial data, develop predictive models, and understand the unique challenges of financial time series, market microstructure, and risk management. The course culminates in a hands-on project simulating a real-world fintech problem, preparing students for roles in quantitative research, data science, and data engineering within financial institutions.

LEARNING OUTCOMES

- Understand the structure and characteristics of financial markets and financial data
- Apply data preprocessing and feature engineering techniques for financial data, including but not limited to time series data
- Implement machine learning models tailored for financial applications
- Evaluate models using appropriate financial metrics and backtesting techniques
- Gain practical experience through a hands-on project involving real financial datasets
- Develop skills relevant for fintech roles: quantitative analyst, data scientist, data engineer, and similars

COURSE LEARNING OBJECTIVES

Learning Objective #1: Introduction to Financial Markets & Data Science Objectives:

- Understand the fundamental structure of capital markets, asset classes, and the roles of various market participants.
- Identify and differentiate between core financial data types, including prices, volumes, order books, and fundamental data.
- Analyze market microstructure and the mechanisms of trading.
- Explore career pathways within the fintech landscape.

Learning Objective #2: Financial Data: Acquisition, Data Structures, Visualization & Preprocessing Objectives:

- Master financial data structures and the manipulation of financial time series.
- Acquire, clean, and preprocess financial datasets for analysis.
- Apply Python libraries (Pandas, NumPy, Matplotlib) to visualize financial trends and patterns.
- Execute a practical case study involving real-world financial data.

Learning Objective #3: Data Labeling, Feature Engineering & Feature Importance Objectives:

- Implement data labeling techniques suitable for supervised learning tasks.
- Construct technical indicators (e.g., Moving Averages, RSI, MACD) and incorporate alternative data features.
- Apply dimensionality reduction techniques, such as Principal Component Analysis (PCA).
- Evaluate feature importance to select the most predictive variables.

Learning Objective #4: Machine Learning Modeling in Finance (Supervised Learning) Objectives:

- Develop and apply linear models to financial prediction tasks.
- Implement ensemble models and deep learning architectures for market analysis.
- Compare various supervised learning approaches in the context of a financial case study.

Learning Objective #5: Model Training, Cross-Validation & Backtesting Objectives:

- Perform rigorous cross-validation and hyperparameter tuning to optimize model performance.
- Design and execute backtesting strategies to validate trading hypotheses.
- Apply Monte-Carlo simulations to assess model robustness and potential outcomes.

Learning Objective #6: Machine Learning Modeling in Finance (Unsupervised & Reinforcement Learning) Objectives:

- Apply unsupervised learning techniques, including clustering, to identify market regimes or asset groupings.
- Understand the fundamentals of Reinforcement Learning in the context of trading agents.
- Utilize dimensionality reduction for complex financial datasets.

Learning Objective #7: Alternative Data & Natural Language Processing (NLP) Objectives:

- Identify sources of alternative data and navigate the challenges of acquisition and integration.
- Apply Natural Language Processing (NLP) techniques to unstructured financial text.
- Evaluate how sentiment analysis and other NLP applications influence financial decision-making.

Learning Objective #8: Financial Derivatives & Fixed Income Objectives:

- Understand the mechanics of financial derivatives, including Futures, Swaps, and Forex.
- Analyze Options pricing strategies and applied models (e.g., Black-Scholes-Merton).
- Examine the characteristics and valuation of fixed-income securities.

Learning Objective #9: Profitability, Risk Management & Portfolio Optimization Objectives:

- Differentiate between Alpha and Beta and calculate profitability ratios like the Sharpe ratio.
- Compute risk metrics, including Value at Risk (VaR) and Conditional Value at Risk (CVaR).
- Apply Modern Portfolio Theory (Mean-Variance Optimization) and hedging strategies.
- Integrate machine learning predictions into the portfolio construction process.

Learning Objective #10: Financial Markets Simulation Objectives:

- Model asset price paths using Geometric Brownian Motion (GBM) and Jump Diffusion models.
- Implement square-root simulations for volatility modeling.
- Identify and avoid common pitfalls in financial market simulations.

Learning Objective #11: Ethical, Regulatory, and Practical Considerations Objectives:

- Navigate data privacy laws and compliance requirements in financial data science.
- Ensure model interpretability and explainability to meet regulatory standards.
- Assess and mitigate bias in financial machine learning models.
- Plan for the practical challenges of deploying models in live fintech environments.

COURSE READINGS AND HANDOUTS

- Course videos, handouts, and additional readings available on the Canvas course site.

COURSE FORMAT AND PROCEDURES

Class Sessions

Our class sessions will be in the format of lectures, guest speakers, hands-on labs & class discussions. Students will additionally prepare and lead brief discussions on pre-defined topics provided by the lecturer.

Canvas Course Site

The Canvas course site is arranged in modules by class number, which include the lecture slides, readings, discussion boards, class handouts, and assignments. Be sure you have your notification preferences set in Canvas to get email notifications of announcements from the course site. If you will access Canvas from a different time zone than Eastern Time, set your time zone in Canvas so you will see the correct due dates and times for assignments.

Use of Devices

Students are encourage to bring a laptop or tablet to class to take notes and follow along the hands-on labs. Please use your devices (laptop, tablet, phone) during class sessions only in ways that support your focus and engagement during class. Please ***do not*** spend class time doing anything on your devices that does not directly relate to staying focused on the work we are doing together during the class session (for example, doing homework for another class, messaging friends, and looking at social media).

Attendance and Completion of Assignments

Attending class, participating in discussions, and completing assignments on time are essential to learn the most from the class, to support the learning of your classmates, and to get timely feedback on your work.

OFFICE HOURS

The purpose of office hours is to have unstructured time when you can ask me questions, continue discussions, ask for feedback, or talk about anything you want to talk about. Meeting with me during office hours is optional, but I encourage you to always feel comfortable meeting with me to ask questions, discuss feedback, or get help with an assignment you are working on. Office hours are by appointment only, on Wednesdays, 5:00—6:00pm, Aidekman Center, Room 002; virtual office hours available as well. Email me beforehand.

COMMUNICATION OUTSIDE OF CLASS OR OFFICE HOURS

If you have questions outside of class and are not able to meet with me during office hours, please send me an email with your questions.

CLASS SCHEDULE AND ASSIGNMENTS

	Date	Lecture	HW	Project
Week 1	January 14th	Lecture 1		
Week 2	January 28th	Lecture 2	Handout HW1	
Week 3	February 4th	Lecture 3		
Week 4	February 11th	Lecture 4	HW1 Due	
Week 5	February 18th	Lecture 5		Project Proposal Due
Week 6	February 25th	Lecture 6	Handout HW2	
Week 7	March 4th	Lecture 7		
Week 8	March 11th	Lecture 8	HW2 Due	
Week 9	March 25th	Lecture 9		Project Check-In
Week 10	April 1st	Lecture 10		
Week 11	April 8th	Lecture 11		
Week 12	April 15th	Presentations (I)		Final Report, Code & Slides Due
Week 13	April 22nd	Presentations (II)		

NOTES: (1) This schedule is subject to change during the semester. (2) If a class session or assignment due date falls on a religious holy day that you observe, please let me know. You do not need to come to class on that day, and you can turn in any assignments due on another date.

Please note that due to capacity constraints, the final project presentations will be held over two days during Weeks 13 and 14. Students will be randomly assigned to present on one of these days. To ensure fairness, all students must submit their presentation slides by the start of class in Week 13. No changes to the slides will be allowed after this deadline.

COURSE GRADING AND ASSESSMENTS

The final grade is determined by a weighted combination of four key components. In-class attendance and participation account for 30% of the total grade. Because this course relies on interactive discussions, case studies, and guest speaker insights, consistent attendance and active engagement in classroom dialogue are essential requirements for success.

In-class attendance and participation account for 30% of the total grade. Because this course relies on interactive discussions, case studies, and guest speaker insights, consistent attendance and active engagement in classroom dialogue are essential. Attendance will be taken for each class. This 30% is calculated as follows:

- **Attendance** (15%): Students will receive the full 15% if they attend at least 12 out of the 14 scheduled lectures. For each lecture missed below the 12-lecture threshold, the grade is reduced by 3% (e.g., attending 11 lectures results in 12%). Please feel free to reach out to the instructor regarding special situations.

- **Student-Led Discussion** (10%): You will be evaluated on your contribution to the student-led discussion sections, which take place during the last 30 minutes of each lecture.

- **In-Class Participation** (5%): Points are awarded for active engagement and contributions to general classroom discussions.

The remaining 70% of the grade focuses on deliverable work:

Students will complete **two Homework Sets**, each weighted at 20% (40% total), which will test the application of financial data structures and modeling concepts covered in the semester.

The course culminates in a **Final Project**, worth 30%, consisting of a written report and a presentation. This capstone requires students to incorporate key curriculum's topics—from data acquisition to model backtesting—into a comprehensive study of a specific financial market problem.

COMMITMENT TO DIVERSITY, EQUITY, AND INCLUSION

In this course, and in my professional and personal life, I value and affirm people from diverse backgrounds and perspectives. The diverse experiences and viewpoints that students bring to this course are a resource and a benefit to everyone in the course. I intend to create a learning environment that recognizes and supports a diversity of perspectives and experiences and that respects and honors your identities (including race, ethnicity, gender, sexuality, ability, age, socioeconomic status, religion, and culture). I make ongoing efforts to learn about diverse perspectives and identities, and I always welcome feedback about ways that I can improve the inclusiveness and effectiveness of the course to best support all students in their learning.

STUDENT MENTAL HEALTH SUPPORT

As a student, you may experience times when personal stressors or emotional difficulties interfere with your academic performance or well-being. The Tufts Counseling and Mental Health Service (CMHS) provides confidential consultation, brief counseling, and urgent care at no cost for all Tufts undergraduates as well as for graduate students who have paid the student health fee. To make an appointment, call 617-627-3360. Please visit the CMHS website to learn more about their services and resources.

Ears for Peers is a student-run, anonymous, and confidential hotline for Tufts students when they need to reach out to someone. Students call or text for many reasons, including difficulties with school, the college transition, and friends/family struggles. Ears for Peers also receives calls and texts regarding sexual assault and self-harm. To contact Ears for Peers, call or text 617-627-3888 or message through their anonymous website.

TUFTS IN-PERSON CLASSROOM HEALTH AND SAFETY POLICY

We must work together as a community to mitigate the risk of spreading illness. The use of masks on campus is optional, but wearing a mask indoors is highly recommended as an effective strategy to prevent the transmission of illness if you feel like you might be getting sick or are recovering from being sick. If you don't feel well and are experiencing symptoms of COVID-19 or flu, please do not come to class. For more information, review current Tufts public health policies and guidance.

ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

Tufts University values the diversity of students, staff, and faculty and recognizes the important contribution each student makes to our unique community. Tufts is committed to providing equal access and support to all qualified students through the provision of reasonable accommodations so that each student may fully participate in the Tufts experience. If a student has a disability that requires reasonable accommodations, they should contact the StAAR Center (Student Accessibility and Academic Resources Center) at StaarCenter@tufts.edu or 617-627-4539 to make an appointment with an accessibility representative to determine appropriate accommodations. Please be aware that accommodations cannot be enacted retroactively, so request accommodations as early as possible.

RELIGIOUS ACCOMMODATIONS

Tufts University faculty, staff, and administration highly value and acknowledge the religious diversity of its student body. Students seeking religious accommodations related to their holy days are encouraged to collaborate with faculty to make arrangements during the first week of each semester. Students seeking additional support may refer to the University Religious Accommodation Policy. The University Chaplaincy is also available to respond to questions on religious observances; their contact information is available here.

ACADEMIC SUPPORT AT THE StAAR CENTER

The StAAR Center (Student Accessibility and Academic Resources Center) offers a variety of resources and services to all students (both undergraduate and graduate) in the School of Arts and Sciences, School of Engineering, School of the Museum of Fine Arts, and Fletcher School; services are free to all enrolled students. Students may make an appointment to attend subject tutoring in a variety of disciplines or meet with an academic coach to hone fundamental academic skills like time management or overcoming procrastination. Students can make an appointment for any of these services by visiting the Tutor Finder or the StAARCenter website.

ACADEMIC INTEGRITY AND PLAGIARISM

Tufts holds its students accountable for adhering to academic integrity policies and meeting the requirements of ethical behavior and academic work as described in the Tufts University policies. Each student is individually responsible for understanding and following these policies. If you have any questions about academic integrity or about the expectations for a particular assignment in this course, be sure to ask me for clarification.

Faculty at Tufts are required to report suspected cases of academic integrity violations to the Office of the Dean of Student Affairs. If I suspect that you have cheated or plagiarized in this course, I am required by the university to report my concerns.

As part of this course, I will use Turnitin on some assignments to help determine the originality of your work. Turnitin is an automated system that instructors can use to compare each student's assignment with all websites and published material available online, as well as a database of all student papers submitted to Turnitin in any course at any participating school during the current and previous years. Turnitin retains a copy of all submitted work in the Turnitin database for the sole purpose of detecting plagiarism in future submitted works. Students retain copyright on their original course work.

POLICY ON USE OF ARTIFICIAL INTELLIGENCE (AI) AND LARGE LANGUAGE MODELS (LLM)

Use of any generative AI tools and LLMs (such as ChatGPT, Google Gemini, Microsoft Copilot, Claude, or other AI tools) is subject to the same Tufts Academic integrity policies that guide all of your coursework at Tufts. Use of AI tools in a way that is helpful for your learning is acceptable, but any use that leads to misrepresentation of your individual knowledge or skills on an assignment is not acceptable and is a violation of Tufts academic integrity policies. If you have questions about the use of AI in this course that are not covered in the syllabus, please ask during class or office hours.

In this course, you may use AI tools and LLMs for your learning, just as you can collaborate with your peers for things such as brainstorming, getting feedback, revising, or editing of your own work. However, you may not submit any work generated by an AI program as your own work. While AI tools and LLMs can be valuable writing assistants, please remember that you are ultimately responsible for all content in your submitted work.

Citation of AI and LLMs

- If you use an LLM (such as ChatGPT, Google Gemini, Microsoft Copilot, and Claude) or any other AI tool to help with writing any of the assignments in this course, you must include it as a source in a list of references for the assignment and also cite it in the text of your paper to acknowledge the specific sections of text for which you used AI and the way you used it.
- Any statement in your work directly generated by an LLM should be in quotation marks, but avoid or minimize the quotation of generated text unless it serves a necessary purpose in relation to the subject of your paper (for example, if the purpose of your paper is to compare the effectiveness of different generative AI tools).
- Approaches to the citation of AI are rapidly evolving. Please see this helpful overview for some different approaches to the citation of AI. Instead of rigidly following a specific citation format, I recommend that you focus on providing clear, easy-to-understand information about the specific way that you used AI tools: for example, you can include a statement that you used an AI tool to identify and correct any grammatical or spelling errors in your writing, or you used AI to get started in thinking about topics for a paper.

Guidelines for Making Decisions about AI Use in this Course

- Familiarize yourself with AI tools and LLMs, not only in relation to the output they can create but also in relation to the way that they were created. Text from LLMs may closely mimic human knowledge, understanding, and even human emotions. Bias is embedded in the creation of AI systems and in their output, and you may encounter harmful or inaccurate language and ideas. LLMs can produce inaccurate or false information with confidence and can invent false reference sources.
- LLMs often cannot produce an accurate summary unless you write multiple prompts based on your own reading and understanding of the text being summarized. Summaries generated by

LLMs are often just a shortening of the original text with the addition of related ideas in ways that may be inaccurate or may not be focused on the most important ideas in the original text. Keep this caution in mind when using AI tools for both reading and writing.

- Be aware of the intellectual property and copyright implications of using AI and LLMs. Many of these tools retain the rights to use your information and content shared with them in a variety of ways.
- Review information from Tufts Technology Services on the costs and risks of using generative AI tools.

Use of Rumi for AI Awareness

We might be using Rumi for some assignments in this course. Rumi AI is a tool that has been adopted by Tufts Technology Services to promote academic honesty in the age of AI.

One primary intended use of Rumi is for students to demonstrate compliance with course policies on the use of AI. Rumi also supports ethical AI use while providing equitable access to LLMs. We will use Rumi not as a tool of compliance but instead as a tool to better understand ways to use AI and better understand your own writing process.

Compose specified assignments directly in the Rumi application. When you open an assignment in Canvas, you will be directed to Rumi if that assignment is supposed to be completed in Rumi. The overall structure and functions in Rumi are similar to Google Docs or Microsoft Word. The following guidelines will also be helpful.

Writing Process Analysis: Rumi analyzes your typing behavior and creates a record of your writing process similar to the version history in Google Docs. You can view your typing and version history to get a better understanding of the process you use when writing, including whether AI is part of your writing process. Your typing history is also visible to the instructor, and Rumi will analyze your typing behavior to flag possible plagiarism or AI misuse. Writing that is flagged is not automatically considered plagiarism or AI misuse, but I may ask questions about how you completed the assignment or your understanding of the work you submitted if I am concerned that your work is not consistent with Tufts academic integrity policies.

Pasted Text Attribution: If you need to copy and paste text from another location, Rumi allows you to attribute the pasted text from external sources by explaining its origin. Typical examples include quotes with proper attribution, outlines created on other devices, or citations generated using citation management software. Do not copy and paste text into Rumi that was generated by an AI tool outside of Rumi.

AI Prompts: Rumi will provide access to pre-approved prompts that can assist you in your writing. You can select from various pre-created prompts that align with the AI use policy in this course: for example, to correct spelling, grammar, or sentence structure.

AI Freeform Use: In some assignments, Rumi allows you to freely engage with multiple AI models, as long as you follow the AI use policy in this course. Rumi currently includes access to Claude, Gemini, and ChatGPT. Your interactions with the LLM will be automatically recorded in Rumi and can be reviewed as part of the assignment evaluation by you, your instructor, and your peers, if you chose to share your assignment during class discussions.

POLICY ON SHARING COURSE MATERIALS

I have tried to design this course for everyone to feel comfortable fully participating in discussion boards, attending class sessions, asking questions, learning, and giving feedback. Any recordings, documents, or other content posted by students or the instructor in the Canvas course site is protected by copyright and will be shared only with the enrolled students in the course and will not be posted publicly. Tufts policy prohibits anyone from sharing any content made available in this course, including but not limited to the course syllabus, reading materials, videos, handouts, and assignments, with anyone outside of the course without permission from the instructor. This policy especially includes any posting or sharing of documents, videos, or audio recordings on publicly accessible websites or forums. Any such posting could violate laws that protect copyright and protect the privacy of student educational records.

Appendix: Capstone Project Guidelines

Project Overview:

The final project gives you the opportunity to apply the concepts learned throughout the course to a real-world financial problems. You will work individually to design, implement, and evaluate a predictive model or trading strategy relevant to financial markets.

Project Goals:

- Demonstrate ability to acquire and preprocess financial data.
- Engineer meaningful features specific to financial applications.
- Develop and tune machine learning models suitable for financial time series or market data.
- Perform robust backtesting and evaluation of your model or strategy.
- Present your methodology, results, and insights clearly in a written report and presentation.

Step-by-Step Project Phases & Deliverables:

Deliverable 1: Project Proposal (Week 5)

- **Deliverable:** 1-page proposal document.
- **Content:**
    - Problem statement: What financial problem are you addressing? (e.g., stock return prediction, credit risk scoring, algorithmic trading strategy)
    - Dataset(s) you plan to use and data sources.
    - Brief description of your planned approach and ML methods.
    - Expected challenges and how you plan to address them.

Instructor Feedback: You will get feedback to refine the scope and feasibility.

Deliverable 2: Final Report, Slides & Code (Week 12)

- **Deliverable:**

    - **Written Report (5-8 pages)** including:
        - Introduction and problem definition.
        - Data description and preprocessing.
        - Feature engineering and modeling approach.
        - Results and evaluation.
        - Discussion of challenges, limitations, and potential improvements.
        - Conclusion.
    - **Slides for Final Presentation**

- **Codebase for the project.** The code is a critical component of the final grade and must be fully executable. To ensure reproducibility, submissions must adhere to the following best practices:
  - Reproducibility: The code must run end-to-end without errors.
  - Dependencies: Include a requirements.txt or environment file listing all necessary libraries.
  - Environment: Clearly specify the Python version used.
  - Data: All datasets required to run the project must be included in the submission (or clear scripts provided to automatically download them).
  - Documentation: Typical software engineering standards should be followed (e.g., clear variable names, comments, and a README explaining how to execute the code).

Deliverable 3: Final Presentation (Weeks 12 & 13)

- **Deliverable:**
  - **Presentation:** 15 minute presentation summarizing your project, methods, and findings, + 10 minutes for questions from instructor & other students

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General Guidelines and Best Practices:

- **Data Sources:** Use credible and reliable financial data sources such as Yahoo Finance, Alpha Vantage, Quandl, or Kaggle datasets. If you use alternative data (social sentiment, news), clearly document sources and processing steps.
- **Reproducibility:** Organize your code and notebooks so others can replicate your results. Use comments and markdown cells to explain your workflow.
- **Ethics and Compliance:** Ensure your project complies with data privacy regulations and ethical standards. Avoid using proprietary or confidential data unless you have permission.
- **Avoid Overfitting:** Use appropriate validation techniques such as walk-forward validation or time-based splits. Be mindful of lookahead bias.
- **Collaboration:** If working in teams, distribute tasks clearly and submit a joint report with individual contributions noted.

Suggested Project Topics:

1. Predicting Next-Day Stock Returns Using Technical and Fundamental Features
2. Developing and Backtesting a Momentum-Based Trading Strategy
3. Credit Risk Classification Using Alternative Data Sources
4. Sentiment Analysis of Financial News or Social Media to Generate Trading Signals
5. Intraday Price Movement Prediction Using Limit Order Book Data
6. Portfolio Optimization Using Machine Learning Forecasts
7. Predicting Stock Returns Using Machine Learning Models
8. Building a Sentiment-Based Trading Signal

9. Developing an Algorithmic Trading Strategy on Intraday Data
10. Credit Risk Modeling with Alternative Data Sources

The list above offers only suggestions. Feel free to explore any topic that interests you, as long as it incorporates concepts discussed in class and you have your instructor's approval. Creativity is always encouraged!

Evaluation Criteria:

- Clarity and relevance of problem statement.
- Quality and appropriateness of data preprocessing and feature engineering.
- Correctness and sophistication of machine learning methods.
- Robustness of model evaluation and backtesting.
- Quality of report and presentation (clarity, insightfulness, professionalism).
- Creativity and critical thinking in addressing challenges.

** Please note that due to capacity constraints, the final project presentations will be held over two days during Weeks 12 and 13. Students will be randomly assigned to present on one of these days. To ensure fairness, all students must submit their presentation slides by the start of class in Week 12. No changes to the slides will be allowed after this deadline.